

Peripherals :

Peripherals are any device that can be attached to a CPU eg. : Hard Disk, mouse, printer.

- peripherals are slow..

They hold up the CPU from carrying out its task.

Interfaces :

What is an Interface?

The interface is the combination of hardware and software needed to link the CPU to the peripherals and to enable them to communicate with the CPU despite all their differing characteristics.

- Hardware is the bit you connect the cable into USB
- Software is the driver disk that you usually need to install when you get the device.

Different characteristics of Computer peripherals

1. Different Data Rate
2. Variety of Code and Control signal
3. Serial and parallel data transmission
4. Even work Higher voltage than the CPU
5. All operate at much slower speeds than CPU

The main function of an interface

1. Buffering
2. Convert data to and from serial and parallel
3. Convert data to and from analog and digital
4. Voltage Conversion
5. protocol conversion
6. Handling status signal.

Buffering :

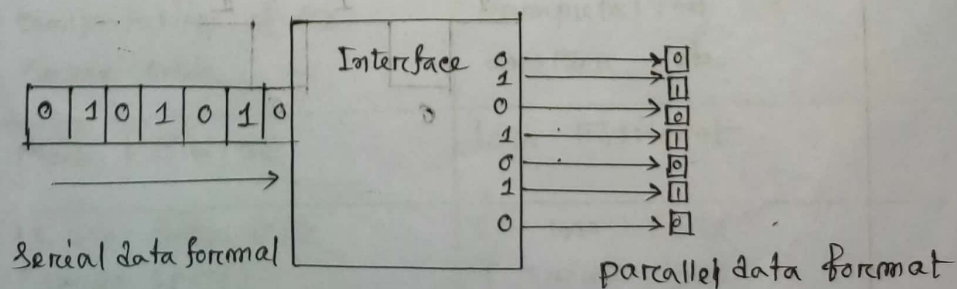
This is the area of RAM within the interface which stores the data while in transmit between the processor and the peripheral.

- use to temporarily store the data
- Balance differences in speed between the peripherals and the CPU.

Serial Data transmission:

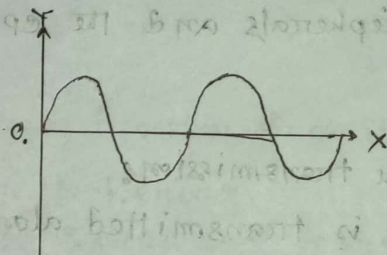
When data is transmitted along a communication channel one bit after another in sequence.

Parallel Data transmission: Transmit several bits of data simultaneously across a series of parallel channels.

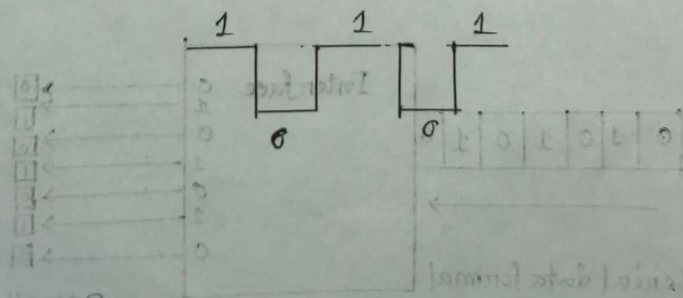


Analogue signal: Many electrical signals are analogue signal that are varying between two limits.

- Analogue signals that are sent in from peripherals to the digital form that the CPU can handle.



Digital signal: Many electrical signals are digital signal that have only two value. High and low or on and off.



A/D and D/A

✓ Analogue to Digital Converter and Digital to Analogue Co.

Interface has to be able to change the digital signal from the computer to an analogue signal that the other device can be understand.

Difference Between Buffering and Spooling.

	Buffering	Spooling
1.	Buffer store data in a memory area temporarily while processing other data	Spooling is a type of multi-programming that allows users to copy data between several device
2.	It needs greater resource management.	It needs less resource management.
3.	It overlaps the I/O of one task with the computation of the same task	It overlaps the I/O of one task with the computation of another job.
4.	More Efficient	Less Efficient
5.	It use Hard Disk (large spool)	It use RAM (small space)

Traditional Computing System : HCI (Human Computer Interaction)

Human Computer Interaction is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them.

Ex: Virtual Reality

Why HCI Important:

1. Effectiveness
2. productivity
3. Morale
4. Safety.

Interfaces

1. Touch as input
2. Motion as input
3. Eye Movement as input
4. Haptic Interfaces
5. Augmented reality:

Augmented reality use the existing real-world environment and puts virtual information.

6. Wearable Computing:

7. Speech input.

Computation device accompany you, rather than you speaking them out.

The Goals of HCI

1. To produce useable and safe system.
2. Understand the factors that determine how people use technology.
3. Develop tools and technique to enable building suitable system.
4. Achieve efficient, effective and safe information
5. put people first.

Seven Design principle of HCI.

1. Equitable use
2. Flexibility in use
3. Simplicity and intuitiveness of use
4. perceptible information
5. Tolerance for errors
6. low physical effort
7. Size and space for approach and use

1. Equitable: Ensure that interface is accessible and usable by all individuals.
2. Flexibility in use: The design accommodates a wide range of individual preferences and abilities.
3. Simple and Intuitive use: use of the design is easy to understand, regardless of the user's experience, knowledge, language skill.
4. Perceptible Information: The design communicates necessary information effectively to the user, regardless of ambient condition.
5. Tolerance for errors: The design minimizes hazards and the adverse consequence of accidental or unintended actions.
6. low physical Efforts: The design can be used efficiently and comfortably and with a minimum of fatigue.
7. Size and Space for Approach and use: Appropriate size and space is provided for approach, reach, manipulation, regardless of user body size and posture.

Norman's Seven principle

1. use both knowledge in world & knowledge in the head.
2. Simplify task structure
3. Make things visible.
4. Get the mapping right.
5. Convert constraints into advantages
6. Design for error
7. When all else fails → standardize

usability:

It is concerned with the making system easy to learn and use.

1. easy to learn
2. easy to use
3. efficient to use
4. safe to use.

Why usability important?

Ex:

we put our document into the photocopier to make 10 copy and stapled. Then we push the big button "c" to start copy.

- Ⓐ The copier makes the copy correctly
- Ⓑ The photocopier settings are cleared and no copies are made.

If we selected Ⓑ we are right.

For this reason usability is important.

The key components of HCI

HCI is made up of four key components

1. The users
2. The goal oriented task
3. The interface
4. The context.

▣ The users: Users are the individuals or groups who involve in the task and their needs and interaction patterns are studied.

▣ The goal oriented task: refers to the objective or goal the user wants to achieve using the computer system. Ex: booking airline for aviation website

▣ The interface: plays a crucial role in enhancing the user interaction experience. Ex: visiting website in SM

▣ The context: The context involves considering the environment and circumstances in which the system is accessed.

context is any information that can be used to characterize the situation of an entity.

Context :

Two types of information held context

1. Auxiliary information : That are not essenti
2. Relevant information : Essential.

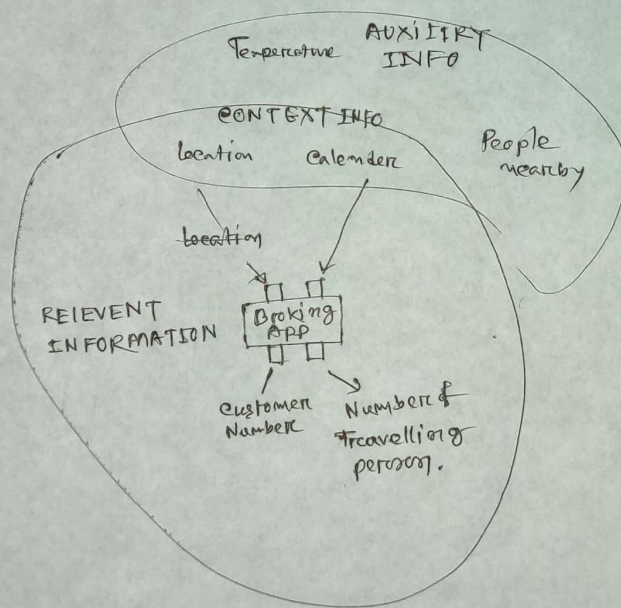


Fig: Context with relative information